

Uncertainty-Guided Latent Diagnostic Trajectory Learning for Sequential Clinical Diagnosis

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Abstract

Clinical diagnosis requires sequential evidence acquisition under uncertainty. However, most Large Language Model (LLM) based diagnostic systems assume fully observed patient information and therefore do not explicitly model how clinical evidence should be sequentially acquired over time. Even when diagnosis is formulated as a sequential decision process, it is still challenging to learn effective diagnostic trajectories. This is because the space of possible evidence-acquisition paths is relatively large, while clinical datasets rarely provide explicit supervision information for desirable diagnostic paths. To this end, we formulate sequential diagnosis as a Latent Diagnostic Trajectory Learning (LDTL) framework based on a planning LLM agent and a diagnostic LLM agent. **For the diagnostic LLM agent, diagnostic action sequences are treated as latent paths and we introduce a posterior distribution that prioritizes trajectories providing more diagnostic information.** The planning LLM agent is then trained to follow this distribution, encouraging coherent diagnostic trajectories that progressively reduce uncertainty. Experiments on the MIMIC-CDM benchmark demonstrate that our proposed LDTL framework outperforms existing baselines in diagnostic accuracy under a sequential clinical diagnosis setting, while requiring fewer diagnostic tests. Furthermore, ablation studies highlight the critical role of trajectory-level posterior alignment in achieving these improvements.

1 Introduction

Accurate and timely diagnosis is a fundamental and sequential process in clinical practice. In real-world clinical settings, when a patient was admitted to a hospital, clinicians seldom can obtain all necessary patient information at once. Instead, they typically start with partially observed information such as symptoms and medical history, and sequentially decide which examinations, laboratory tests,

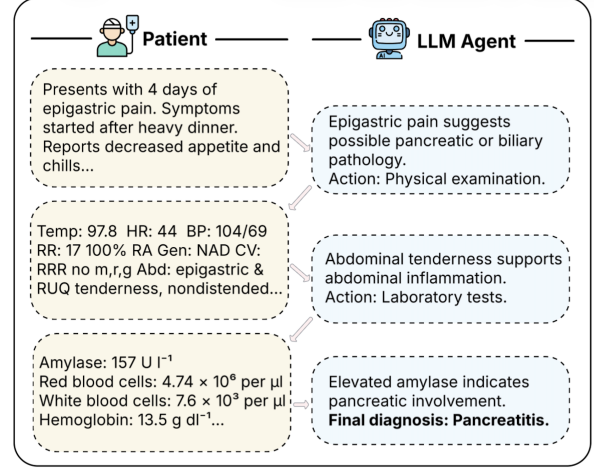


Figure 1: An real-world example of the sequential clinical diagnosis system.

or imaging studies to order in the next step. Each newly acquired result will help refine the clinicians' hypothesis and progressively reduces uncertainty until sufficient confidence is reached for a final diagnosis decision. A concrete real-world example of the sequential clinical diagnosis system is shown in Figure 1. Throughout this process, physicians must balance diagnostic accuracy with constraints of time, cost, and patient burden, making test selection a key part of clinical decision-making (Rajpurkar et al., 2022).

Recent advances in LLMs have demonstrated their strong capabilities in medical reasoning and diagnosis (McDuff et al., 2025; Brown et al., 2020; Achiam et al., 2023; Singhal et al., 2023b). However, most existing approaches assume a full information setting in which all structured and unstructured patient data (including test results) are available and accessible beforehand (Singhal et al., 2023a; Gilson et al., 2023; Achiam et al., 2023). This assumption, however, is impractical due to the aforementioned real-world sequential clinical diagnosis process in which clinicians need to determine which tests to be ordered before making a confident diagnosis. Recent benchmarks such

as MIMIC-CDM (Hager et al., 2024) and interactive diagnostic frameworks like MediQ (Li et al., 2024) begin to model stepwise evidence acquisition, but LLM-based systems still struggle to faithfully reflect practical diagnostic paths without explicit modeling the test-selection process.

A natural solution is to frame diagnosis as an iterative agentic process, where a model sequentially selects diagnostic actions and updates its predictions. This perspective is aligned with recent LLM agent frameworks that enable multi-step reasoning and action selection (Yao et al., 2022; Shinn et al., 2023; Wei et al., 2022; Schick et al., 2023; Yao et al., 2023; Wang et al., 2024). While reinforcement learning has been explored for cost-sensitive test selection (Yu et al., 2023; Sun et al., 2024; Schulman et al., 2017) and hypothesis-driven clinical agents (Bani-Harouni et al., 2025b), reward-driven optimization alone does not provide explicit control over the structure of diagnostic trajectories. In particular, learning purely from terminal rewards does not regulate how diagnostic paths are distributed in the latent reasoning space, and can be unstable in settings where only limited clinical trajectories are available for training.

To this end, we present a Latent Diagnostic Trajectory Learning (LDTL) framework for sequential clinical diagnosis. Instead of directly optimizing test-selection policies solely via reward signals, we treat diagnostic test paths as latent variables and explicitly model their role in reducing diagnostic uncertainty. Our proposed LDTL framework consists of two key components: a planning LLM agent that selects the next clinical action based on the current information, and a diagnostic LLM agent that predicts the disease label and produces a calibrated confidence estimate after each step. Diagnosis can terminate early once sufficient confidence is reached. In summary, the main contributions include:

- We develop a sequential diagnosis system in which a planning LLM agent selects diagnostic actions while a diagnostic LLM agent continuously updates the predicted label and confidence, enabling adaptive evidence acquisition and early termination once sufficient certainty is reached. This design improves diagnostic efficiency by avoiding unnecessary tests while maintaining predictive accuracy.
- We formulate diagnostic test selection as a Latent Diagnostic Trajectory Learning (LDTL) problem

and leverage information gain to construct supervision signals based on diagnostic improvement, guiding the planning LLM agent to pursue information-efficient diagnostic paths.

- Based on the real-world abdominal diagnosis benchmark MIMIC-CDM (Hager et al., 2024), our proposed LDTL framework outperform existing baselines while substantially reducing the number of required tests.

2 Related Work

Recent work has increasingly recognized that evaluating medical language models under full information settings fails to capture the sequential nature of real-world diagnosis. (Hager et al., 2024) introduce MIMIC-CDM, a benchmark that simulates stepwise evidence acquisition in abdominal diagnosis. (Li et al., 2024) propose MediQ, framing clinical reasoning as interactive question asking under incomplete information. Other efforts explore multi-agent or consultation style systems for clinical decision-making (Liu et al., 2024; Chen et al., 2025), emphasizing sequential reasoning and structured collaboration. While these approaches model diagnosis as an interactive process, they do not explicitly regularize the distribution of diagnostic trajectories.

In parallel, medical test selection has been studied as a cost sensitive sequential decision problem. Classical reinforcement learning methods (Mnih et al., 2015) have been applied to optimize diagnostic performance under resource constraints (Komorowski et al., 2018; Yu et al., 2023; Sun et al., 2024), and more recent works investigate RL-based confidence calibration and hypothesis-driven clinical agents (Bani-Harouni et al., 2025a,b). These methods primarily rely on reward optimization or cost and accuracy trade-offs, without imposing structural constraints over latent diagnostic paths.

More broadly, general LLM agent frameworks such as ReAct (Yao et al., 2022), Reflexion (Shinn et al., 2023), and Chain-of-Thought prompting (Wei et al., 2022) demonstrate the effectiveness of multi-step reasoning and tool use (Schick et al., 2023; Yao et al., 2023). However, these approaches focus on improving reasoning capability rather than modeling how uncertainty influences test selection in clinical workflows (Guo et al., 2017; Geifman and El-Yaniv, 2017, 2019). Finally, variational latent variable modeling (Kingma and Welling, 2014; Rezende et al., 2014; Blei et al., 2017) provides

principled tools for structured latent representations, but has not been explicitly connected to sequential diagnostic planning (Chung et al., 2015).

3 Methodology

3.1 Problem Statement

We formulate this clinical diagnosis as a sequential decision-making problem under partially observed information. For each patient, the initial observation is an incomplete clinical history h_0 , including symptoms and background information. At each step, the model may select a diagnostic action $a_t \in \mathcal{A}$, where $\mathcal{A}$ denotes the set of available tests or examinations. Executing a_t reveals new clinical evidence, which is incorporated into the patient state h_{t+1} , consisting of the initial history together with all acquired test results until t diagnostic steps. The process terminates at a patient-specific stopping time T , when the model decides that sufficient evidence has been achieved for a final diagnosis.

A complete diagnostic process therefore induces a trajectory $z = (a_1, \dots, a_T)$, and yields a disease prediction $\hat{y}$ for the ground-truth label y . **The goal is to learn a policy that achieves accurate diagnosis while minimizing unnecessary test acquisition.** This requires balancing predictive performance and diagnostic efficiency under uncertainty, without knowing optimal diagnostic trajectories in training.

Unlike static diagnosis settings, where the full patient profile is observed upfront, sequential diagnosis requires the model to actively determine what information to acquire and when to stop. **Accordingly, our formulation emphasizes learning over distributions of diagnostic trajectories rather than optimizing final prediction accuracy alone.**

3.2 LDTL Framework

The details of Latent Diagnostic Trajectory Learning (LDTL) framework are illustrated in Figure 2, we instantiate the diagnostic policy using two collaborative LLM agents: a planning LLM agent and a diagnostic LLM agent. **The planning LLM agent selects the next diagnostic action based on the current patient state, while the diagnostic LLM agent evaluates the updated evidence to produce a disease prediction and a termination decision.**

At step t , given the current state h_t , the planning LLM agent chooses an action $a_t \in \mathcal{A}$. After the corresponding test result is observed, the patient state is updated to h_{t+1} . Conditioned on h_{t+1} , the diagnostic LLM agent predicts a disease label and

determines whether the accumulated evidence is sufficient to stop. If not, the updated state is passed back to the planning LLM agent to produce the next round of action.

Through this sequential process, the two LLM agents generate a patient-specific diagnostic trajectory $z = (a_1, \dots, a_T)$. In next subsection, we will introduce our learning objective for explicitly modeling the distribution over such trajectories.

3.3 Diagnostic Trajectory Modeling

The diagnostic process results in a sequence of information-acquisition actions determined by the planner. We treat this action sequence as a latent diagnostic trajectory that connects the initial patient state to the final diagnostic outcome (Chung et al., 2015). Let h_0 denote the initial patient history and y the ground truth disease label. Executing a trajectory sequentially reveals diagnostic evidence and updates the patient state according to $h_{t+1} = \text{Update}(h_t, a_t)$, where the update operator incorporates the newly observed test result into the accumulated patient context.

The planner models a distribution over possible diagnostic trajectories through its conditional action policy:

$$p_\theta(z \mid h_0) = \prod_{t=1}^T \pi_\theta(a_t \mid h_t), \quad (1)$$

where $\pi_\theta(a_t \mid h_t)$ denotes the probability of selecting action a_t given the current patient state h_t .

Given a trajectory z , the diagnoser evaluates the accumulated evidence and produces a disease prediction. This interaction defines a diagnostic likelihood $p_\psi(y \mid h_0, z)$, which measures how well the collected diagnostic evidence supports the correct disease label.

Under this formulation, the diagnostic process can be viewed as reasoning over a distribution of possible trajectories connecting the initial patient state to the final prediction. Different trajectories may provide varying diagnostic evidence, depending on the sequence of tests selected and the information they reveal. This perspective suggests that diagnostic trajectories should not be treated equally, and those providing stronger evidence for the correct disease should receive higher preference.

3.4 Constructing the Trajectory Posterior

Based on the above rationale, we introduce a trajectory-level preference distribution conditioned

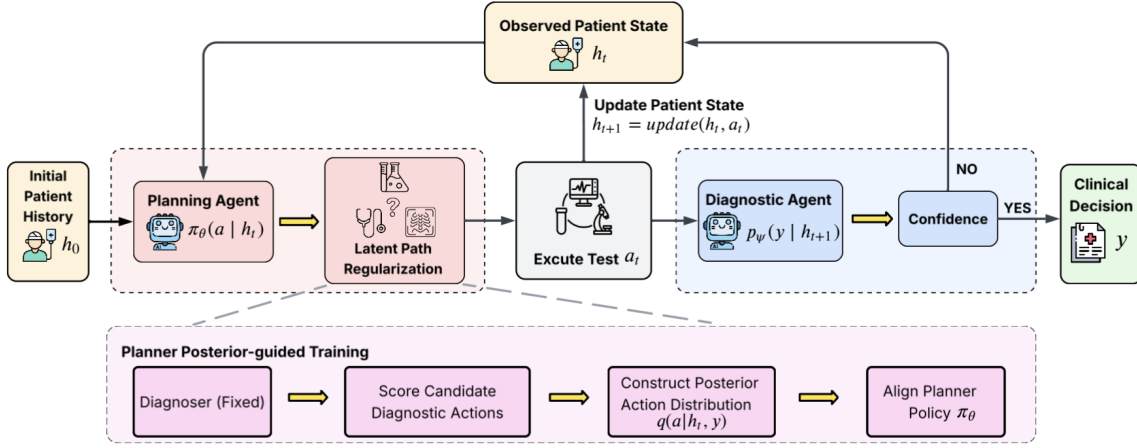


Figure 2: The proposed LDTL framework within an sequential diagnosis system. A planning LLM agent sequentially selects diagnostic tests and a diagnostic LLM agent updates the disease prediction and decides whether to stop based on the accumulated patient and test information.

on the correct diagnosis to guide the planning LLM agent to pursue diagnostically informative action sequences. Intuitively, diagnostic trajectories that provide stronger evidence for the true disease should be assigned higher probability.

Formally, we define a trajectory preference distribution $q(z | h_0, y)$ over candidate diagnostic trajectories as an energy-based model (Ziebart et al., 2008):

$$q(z | h_0, y) = \frac{\exp(\beta S(z))}{\sum_{z' \in \mathcal{Z}(h_0)} \exp(\beta S(z'))}, \quad (2)$$

where $S(z)$ measures the diagnostic informativeness of trajectory z , reflecting how much the sequence of diagnostic actions improves confidence in the correct disease. In practice, directly computing $S(z)$ for full trajectories is intractable because the trajectory space grows exponentially with the planning horizon. β controls the sharpness of the distribution, and $\mathcal{Z}(h_0)$ denotes the set of feasible trajectories starting from the initial patient state h_0 . The score $S(z)$ reflects how informative the diagnostic trajectory is for identifying the correct disease. Trajectories that lead to more confident and accurate predictions are therefore assigned higher preference. This construction encodes trajectory-level structural guidance without requiring explicit supervision of optimal diagnostic paths.

3.5 Action-Level Posterior

Although trajectory preferences capture global diagnostic information across an entire action sequence, directly optimizing over the trajectory space is computationally intractable as the number of possible trajectories grows exponentially with

the decision horizon, making explicit reasoning over the trajectory space impractical. Diagnostic trajectories are generated sequentially, where each action depends only on the current patient state. This property allows us to marginalize the trajectory preference distribution into a step-wise action posterior.

Specifically, at the step t , the posterior probability of selecting action a_t can be obtained by summing the probabilities of trajectories whose t -th action equals a_t :

$$q(a_t | h_t, y) = \sum_{z: z_t = a_t} q(z | h_0, y), \quad (3)$$

which converts trajectory-level preferences into action-level supervision that can be applied locally at each decision step.

In practice, directly computing the trajectory score $S(z)$ is intractable due to the exponential trajectory space. We therefore approximate the induced action posterior using step-wise diagnostic improvement signals derived from the diagnostic LLM agent. Given the current patient state h_t , the diagnostic LLM agent produces a predictive distribution over diseases. For a candidate action a , we evaluate its diagnostic utility using information gain (Settles, 2010; MacKay, 1992):

$$IG(h_t, a) = \log p(y | h_{t+1}) - \log p(y | h_t), \quad (4)$$

where $h_{t+1} = \text{Update}(h_t, a)$ incorporates the newly observed test result.

Actions that increase the model’s confidence in the correct diagnosis receive higher scores. Therefore, we define the action-level posterior as

$$q(a | h_t, y) \propto \exp(IG(h_t, a)/\tau), \quad (5)$$

where τ is a temperature parameter controlling distribution smoothness. **This construction enables efficient inference by replacing exponential trajectory enumeration with evaluation over the finite action set at each step, while still preserving trajectory-level preferences through recursive state updates.**

3.6 Learning Strategy

Training the planning LLM agent and diagnostic LLM agent jointly often leads inferior performance due to unstable optimization, as both components depend on evolving predictions of the other (Ouyang et al., 2022). To decouple diagnostic prediction from action learning, we adopt a two-stage learning strategy.

Stage 1: Fine-tune diagnostic LLM agent. We first fine-tuning the diagnostic LLM agent using supervised information with full patient information. **Given the complete set of diagnostic results, the model learns to predict the correct disease label and produce a confidence score reflecting predictive certainty.** This stage serves two purposes. First, it stabilizes the output format and probability distribution of the LLM for clinical diagnosis tasks. Second, it provides a reliable diagnostic reference model whose predictions can be evaluated under different information states. After convergence, the diagnostic LLM agent is frozen and used as a fixed module for planning LLM agent learning.

Stage 2: Planner LLM learning. By fixing the diagnostic LLM agent, we train the planning LLM agent to select diagnostic actions sequentially. The planner LLM induces a trajectory distribution through its conditional action policy as defined in Eq. 1, where h_t denotes the patient state after incorporating previously observed test results. Using the action-level posterior $q(a_t | h_t, y)$ constructed in Eq. 5, we convert trajectory-level diagnostic preferences into step-wise supervision. The planner LLM is optimized by aligning its policy with this posterior through

$$\mathcal{L}_{\text{planner}} = \sum_{t=1}^T \text{KL}(q(a | h_t, y) \parallel \pi_{\theta}(a | h_t)), \quad (6)$$

which encourages the planner to align its action distribution with the posterior derived from diagnostic improvement. Although optimization is performed locally at each step, the posterior $q(a | h_t, y)$ aggregates information from trajectory level preferences, thereby imposing structural regularization over diagnostic paths.

Through this two-stage training strategy, the diagnostic LLM agent provides a fixed probabilistic reference model, while the planning LLM agent learns to select informative diagnostic actions. During training, the action posterior $q(a_t | h_t, y)$ is computed using the ground-truth diagnosis y to provide supervision for the planning LLM agent. During inference time, the planning LLM agent selects actions directly according to its learned policy $\pi_{\theta}(a_t | h_t)$ without access to the true label.

4 Experiments

In this section, we describe the experimental setup and evaluate LDTL through comparisons with baseline methods, ablation studies, efficiency analysis, and case studies.

4.1 Experimental Setup

4.1.1 Dataset and Pre-processing

We evaluate on MIMIC-CDM (Hager et al., 2024; Goldberger et al., 2000), a curated subset of MIMIC-IV (Johnson et al., 2023) designed to support sequential clinical decision-making. The benchmark contains 2400 inpatient cases labeled with one of four abdominal conditions: appendicitis, cholecystitis, diverticulitis, pancreatitis. Each case contains multiple types of clinical data. The initial record includes patient history information (documented symptoms, comorbidities, and family history). In addition, diagnostic tests also provide physical examination notes, imaging reports covering CT, X-ray, ultrasound, and MRI modalities, as well as laboratory records spanning blood, urine, and microbiology tests. But not every diagnostic test is available for every patient. Details of the data preprocessing procedure and action-space construction are described in Appendix B.

4.1.2 Evaluation Metrics

We evaluate both diagnostic performance and decision efficiency. Diagnostic performance is measured using final prediction accuracy and F1 score at termination, where termination is defined as either make a diagnosis or reaching a predefined maximum number of decision steps. To assess efficiency, we report the number of diagnostic tests required per case and the proportion of cases that terminate before reaching the maximum planning steps. These metrics allow us to examine the trade-off between diagnostic accuracy and information acquisition cost.

Table 1: Performance comparison of our method LDTL and baseline methods. We report class-wise accuracies and F1-scores. ZS = zero-shot.

Method	Accuracy				F1-Score	
	Appendicitis	Cholecystitis	Diverticulitis	Pancreatitis	Mean	
SFT-all	97.9	93.1	90.4	90.7	94.2	95.8
Fixed info (history) (ZS)	47.9	34.6	73.1	79.6	54.1	58.8
Fixed info (history + one test) (ZS)	50.5	47.7	76.9	72.2	58.3	60.5
Fixed info (history + two tests) (ZS)	60.9	60.7	78.8	80.5	67.2	68.2
All info (ZS)	73.9	70.0	82.7	87.9	76.9	76.7
Planner random selection	82.8	85.4	90.4	85.2	84.8	83.7
ReAct (Yao et al., 2022)	90.2	79.7	66.7	62.9	74.9	79.1
LA-CDM (Bani-Harouni et al., 2025b)	<u>93.1</u>	83.6	75.0	73.5	81.3	<u>84.1</u>
LDTL	98.9	95.4	<u>78.8</u>	87.9	93.4	91.7

4.1.3 Baseline Model

We use Llama-3-8B (Grattafiori et al., 2024) as the backbone model due to its strong general reasoning capability and open source availability. The 8B scale provides a practical balance between reasoning performance and computational efficiency. To the best of our knowledge, the model has not been pretrained on the MIMIC-CDM dataset. All experiments are conducted using publicly available pretrained weights without any additional pretraining on the dataset. Based on this backbone, we compare our method against several representative baselines that differ in how diagnostic information is acquired and utilized.

One-shot full-information model. All available clinical evidence is provided to the base model at once, and a diagnosis is predicted without sequential decision making. This setting evaluates performance under complete information access.

Static information strategies. Instead of adaptive test selection, we use fixed information subsets formed by combining the patient history with a predefined set of test categories. Specifically, we consider (i) history only, (ii) history plus exactly one test category (physical examination, laboratory tests, or imaging), and (iii) history plus two of the three test categories. The model receives these inputs in a single forward pass and produces a diagnosis. These baselines assess performance under non-adaptive but structured information acquisition.

Random policy. At each decision step, a diagnostic action is sampled uniformly from the action space until termination. This baseline isolates the benefit of learned planning compared to uninformed exploration.

Sequential agent baselines. We further compare with prior language agent decision frameworks,

including the prompting based agent ReAct (Yao et al., 2022) and the reinforcement learning based method LA-CDM (Bani-Harouni et al., 2025b).

4.2 Main Results

Table 3 presents the diagnostic performance of our method compared with a various set of static and sequential baselines. We report class-wise accuracies, mean accuracy, and F1 score.

We first consider the SFT-all setting, where the diagnostic LLM agent is fine-tuned and evaluated with access to complete patient information. This setting represents an approximate upper bound, achieving 94.2% mean accuracy and 95.8 F1 score, since it eliminates the need for sequential test selection and operates under complete information.

Under zero-shot static strategies, diagnostic performance drops substantially when information is limited. Using history only yields 54.1% mean accuracy. Adding one test improves performance to 58.3%, and adding two tests further increases it to 67.2%. Providing all information without training reaches 76.9% mean accuracy, but still remains well below the SFT-all setting. These results indicate that simply exposing the base model to more clinical evidence does not close the gap without task-specific supervision.

We next evaluate sequential decision frameworks. A random planner that selects actions uniformly at each step achieves 84.8% mean accuracy. The improvement over static baselines is not due to the amount of information provided, but to the difference in inference structure. Static strategies approximate diagnosis as a single-step posterior estimation over a fixed evidence set. In contrast, the sequential framework applies a recursive update process in which the posterior at step t becomes the implicit prior for step $t + 1$, and predictions are conditioned on a progressively expanding informa-

tion state. This recursive conditioning reshapes the inference dynamics. Rather than aggregating heterogeneous evidence in a single forward pass, the model performs intermediate belief updates before incorporating additional observations. Such staged evidence integration can reduce uncertainty accumulation and improve calibration stability. However, because the planner does not optimize action selection, the acquired information may be suboptimal, which limits overall performance compared with structured planning strategies.

We next compare against prior sequential decision-making agents under the same test-request interaction loop. ReAct is a prompting-based, zero-shot baseline, whereas LA-CDM optimizes the agent via task-specific training objectives and learns to couple hypothesis refinement with action selection. Consistent with this design, ReAct underperforms trained LA-CDM in mean accuracy, and the gap reflects the benefit of explicit optimization for clinical decision-making rather than the use of an iterative interface per se. In addition, LA-CDM incorporates dedicated training signals for hypothesis quality and decision efficiency, which improves performance compared with its untrained counterpart and other prompting-only variants. However, LA-CDM primarily optimizes intermediate hypothesis quality and decision efficiency through step-wise objectives, and does not directly align the sequential decision process with the final diagnostic outcome. This objective mismatch may lead to locally reasonable actions that do not consistently improve final prediction accuracy.

Our method achieves 93.4% mean accuracy and 91.7 F1 score, outperforming other baselines and approaching the SFT-all upper bound while operating under sequential information constraints. The improvement is consistent across most disease categories, with obvious gains in appendicitis and cholecystitis. These results suggest that aligning the step-wise decision process with the final diagnostic objective leads to more effective information acquisition. By explicitly coupling intermediate action selection with outcome level supervision, the model learns to prioritize tests that contribute directly to improving final prediction accuracy rather than merely refining intermediate hypotheses.

Overall, the results demonstrate that structured test planning is critical for reliable clinical decision-making. Static aggregation of information is insufficient, and naive action selection degrades performance, whereas trajectory-aware planning substan-

tially improves prediction accuracy.

4.3 Ablation Study

To evaluate the effect of latent path regularization, we compare three variants: (1) random selection (RS), which samples actions uniformly from the action space at each step; (2) a state conditioned planner trained without latent path regularization (w/o LP), which selects actions conditioned on the current state but does not incorporate trajectory-level posterior aggregation to enforce consistency across steps; and (3) our full model with trajectory-level posterior alignment.

Table 2: Ablation study on Llama3-8B. We compare random selection (RS), a state-conditioned planner without latent path regularization (w/o LP), and our full model with trajectory-level posterior alignment.

Method	Acc	Macro-F1	App.	Chol.	Div.	Pan.
RS	84.8	83.7	82.8	85.4	90.4	85.2
w/o LP	88.6	88.5	94.8	90.7	80.7	78.7
LDTL	93.4	91.7	98.9	95.4	78.8	87.9

As shown in Table 2, the three variants exhibit a clear performance hierarchy. Random selection (RS) yields the lowest accuracy 84.8%, as actions are sampled uniformly without conditioning on the current patient state. This baseline does not learn a policy and therefore cannot adapt information acquisition to the diagnostic context.

Training a state conditioned planner without latent path regularization (w/o LP) improves accuracy to 88.6%. This indicates that learning a local policy $\pi_{\theta}(a | h_t)$ already provides meaningful gains over random exploration by selecting actions based on the current information state. However, this variant optimizes action selection only at the step level, without enforcing consistency over entire diagnostic trajectories. Thus the planner tends to select less diagnostically informative evidence sequences, resulting in inferior final predictions.

In contrast, our full model LDTL improves accuracy to 93.4% and F1 to 91.7. Unlike w/o LP, it aligns the planner with a trajectory-aware posterior derived from diagnostic improvement, aggregating trajectory-level preferences into action-level supervision that regularizes the distribution of diagnostic paths beyond standard state conditioned policy learning.

4.4 Diagnostic Efficiency Analysis

Beyond final accuracy, we analyze how many diagnostic steps are required before termination across the test dataset. Figure 4 reports the number of

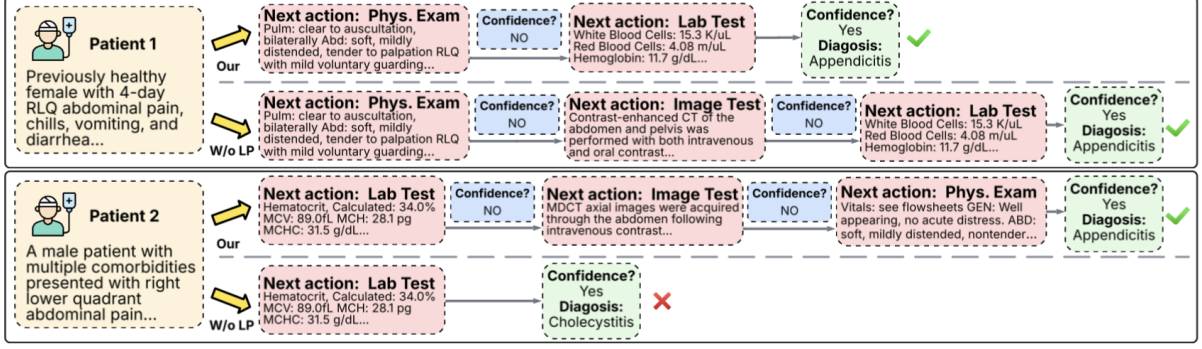


Figure 3: Case studies comparing our method LDTL with the variant without latent path regularization (w/o LP).

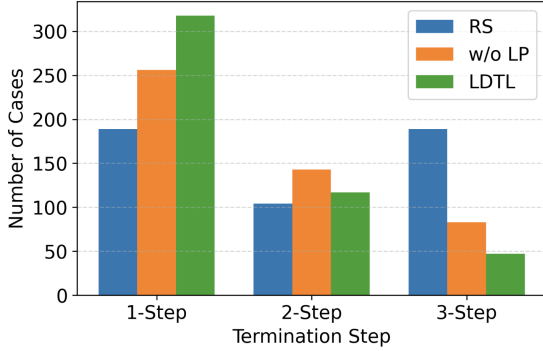


Figure 4: Distribution of diagnostic termination steps on the test cases. Numbers indicate how many cases terminate after one, two, or three actions.

cases that terminate after one, two, or three actions for each method.

Our method achieves the highest proportion of one-step terminations while requiring the fewest three-step trajectories. This indicates that our model (LDTL) can more frequently identify decisive diagnostic evidence early and avoid unnecessary additional tests.

In contrast, random selection exhibits a large number of three-step trajectories, reflecting inefficient action choices that fail to quickly reduce diagnostic uncertainty. The state conditioned planner without latent path regularization improves over random exploration but still requires more multi-step trajectories compared to our approach.

These results suggest that trajectory level posterior alignment helps the planner select more informative diagnostic actions earlier in the process, enabling faster convergence to confident decisions.

4.5 Case Study

To further understand the behavioral differences resulting from latent path regularization, we analyze representative diagnostic trajectories from the test set in Figure 3.

Efficient evidence acquisition. In Patient 1 with true label appendicitis, LDTL first performs a phys-

ical examination and then requests laboratory tests, after which the diagnostic LLM agent reaches sufficient confidence and terminates with the correct diagnosis. In contrast, the planning LLM agent without latent path regularization follows a less efficient trajectory, requesting an additional imaging test before obtaining the same laboratory evidence. This example illustrates that trajectory-level regularization helps prioritize diagnostically informative actions and avoids unnecessary tests.

Avoiding premature termination. Patient 2 demonstrates a different failure mode. LDTL gradually collects evidence through laboratory tests, imaging, and physical examination before reaching a confident and correct diagnosis. Without latent path regularization, however, the planning LLM agent terminates after the first laboratory test and produces an incorrect prediction cholecystitis. This example shows that our method encourages more reliable evidence accumulation and prevents over-confident early decisions.

5 Conclusion

We propose an uncertainty-guided latent diagnostic trajectory learning framework for sequential clinical diagnosis that treats test selection trajectories as structured latent variables. By aligning the planning LLM agent with a posterior distribution derived from diagnostic improvement via a diagnostic LLM agent, our proposed LDTL enforces trajectory level consistency while preserving step-wise supervision. Experiments on MIMIC-CDM show that our approach improves both diagnostic accuracy and decision efficiency compared to other baselines. These findings demonstrate that trajectory-aware regularization is critical for stable and effective clinical planning under partial information.

Limitations

Although our framework improves diagnostic decision-making in a simulated setting and obtains encouraging experimental results, it should not be used for real clinical decision support without extensive validation. Large language models may produce incorrect or misleading outputs, and clinical deployment requires strict human oversight. Our study has several limitations as well.

First, experiments are conducted on a single abdominal diagnosis benchmark with a limited number of disease classes and moderate dataset size. Although MIMIC-CDM simulates sequential evidence acquisition, it does not fully reflect the scale, diversity, and noise characteristics of real hospital environments. Evaluating the proposed sequential framework on broader clinical domains datasets remains an important direction for future work.

Besides, our current formulation defines the planner action space at the level of clinically meaningful test categories. This abstraction simplifies sequential decision modeling and ensures stable learning, but does not capture finer-grained decisions at the individual test level within each category. Extending the framework to more granular and heterogeneous action spaces may require additional mechanisms for structured action representation.

Finally, our framework assumes a fixed and pre-defined set of candidate diagnostic actions. In practice, clinical workflows may involve dynamically generated hypotheses, conditional test dependencies, and external knowledge retrieval. Incorporating open ended action generation and structured medical guidelines into the latent path formulation remains an important direction for future research(Shafer and Vovk, 2008; Vovk et al., 2005).

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A Prompts

A.1 Planning Agent Prompt

Planner Prompt

You are an experienced emergency physician responsible for evaluating a patient who presents with abdominal symptoms. Your goal is to plan the diagnostic process by deciding what type of examination should be performed next.

At each step, you will receive the patient's currently available clinical information, which may include symptoms, medical history, previously completed examinations, and their results. Based on this information, determine the most appropriate next diagnostic examination that would help clarify the patient's condition and reduce diagnostic uncertainty.

Your objective is to gather information that progressively narrows down the possible diagnoses and supports accurate clinical decision-making.

Possible diagnostic examinations include:

- laboratory tests
- physical examination
- imaging studies

When selecting the next step, consider what additional information would be most informative for distinguishing between possible diagnoses and guiding further clinical decisions. Avoid repeating examinations that have already been performed.

Examples:

Example 1

Patient information: A 45-year-old male presents with severe abdominal pain and fever.

Clinical consideration: Initial symptoms suggest an acute abdominal process, and physical examination is needed to further localize tenderness and assess severity.

Next diagnostic step: ["physical"]

Example 2

Patient information: A 60-year-old male presents with chronic cough. Laboratory tests show normal CBC. Physical examination reveals wheezing. Completed examinations: ["lab", "physical"]

Clinical consideration: Since basic laboratory and physical findings are already available, imaging would provide additional evidence to further evaluate the underlying cause.

Next diagnostic step: ["image"]

Now evaluate the following patient case.

Patient current information: {input}

Return the next diagnostic step using the following format:

["lab"], ["physical"], or ["image"]

A.2 Diagnostic Agent Prompt

Diagnoser Prompt

You are an experienced emergency physician responsible for determining the most likely diagnosis for a patient presenting with abdominal symptoms.

You will be provided with the patient's clinical information, which may include the patient's history, symptoms, and results from previously completed diagnostic examinations. Consider all currently available clinical evidence before making the decision.

Based on the available evidence, identify the most likely diagnosis among the following conditions:

- appendicitis
- cholecystitis
- diverticulitis
- pancreatitis

In addition, assess whether the currently available information is sufficient to support a confident clinical judgment. If the evidence is adequate to support a reliable diagnosis, indicate that you are confident. Otherwise, indicate that additional diagnostic information may still be needed.

Example:

Patient history: A 45-year-old male presents with severe abdominal pain and fever.

Laboratory results: Elevated white blood cell count.

Imaging results: CT scan shows an inflamed appendix.

Clinical assessment: The combination of abdominal pain, fever, leukocytosis, and imaging evidence of an inflamed appendix strongly supports appendicitis, and the current findings are sufficient for diagnosis.

Answer:

Diagnosis: appendicitis

Confident: YES

Now evaluate the following patient case.
 Patient description:{current patient information}
 Return your answer in the following format:
 Diagnosis: <appendicitis | cholecystitis | diverticulitis | pancreatitis> Confident: <YES | NO>

B Dataset Preprocessing and Training Details

We pre-process the dataset to preserve the benchmark’s sequential semantics while standardizing inputs for LLM-based agents. First, when repeated measurements of the same clinical test were available within a single hospitalization, we retained only the earliest recorded value to approximate a realistic early stage diagnostic setting. Second, MIMIC-CDM provides standardized mappings of test names across patients, which enables consistent action definitions in our decision framework. This normalization is essential for modeling test selection, as inconsistencies in clinical documentation would otherwise prevent consistent querying of the same test across different cases. (Bani-Harouni et al., 2025b) Third, we define the planning action space as a set of clinically meaningful test categories, including physical examination, imaging studies, and laboratory tests. When an action is selected, its associated results are appended to the current patient context in a structured textual format. To the best of our knowledge, MIMIC-CDM is currently the only publicly accessible dataset that supports simulation of hypothesis-driven sequential clinical decision making. We adopt the official patient-level split, using 70% of cases for training, 10% for validation, and 20% for testing.

We adopt Llama-3 8B as the backbone language model and perform parameter-efficient fine-tuning using LoRA. The LoRA rank is set to 16 with a scaling factor of 32. We fine-tune the models using the AdamW optimizer with a learning rate of 2×10^{-5} . The trajectory preference distribution in Eq. (2) introduces a temperature parameter β that controls the sharpness of the posterior distribution; in our experiments we set $\beta = 1.0$. All experiments are conducted on NVIDIA H100 GPUs. Training typically requires approximately 6 hours on two H100 GPUs.

C Additional Model Comparisons

Table 3: Performance comparison of our proposed LDTL with additional business models. (For proprietary LLMs, we perform evaluation through inference-only API calls without using the data for model training, fine-tuning, or retention. This protocol follows the PhysioNet data use agreement and ensures that no protected clinical data are incorporated into external model training.)

Method	Accuracy				F1-Score	
	Appendicitis	Cholecystitis	Diverticulitis	Pancreatitis	Mean	
SFT-all (llama-3-8b)	97.9	93.1	90.4	90.7	94.2	95.8
gpt-4o (OpenAI, 2024b)	99.4	98.4	88.4	88.8	95.6	95.7
gpt-4o-mini (OpenAI, 2024a)	94.7	91.5	82.6	86.1	90.5	90.6
claude-sonnet-4 (Anthropic, 2025)	99.3	98.4	90.3	88.8	95.8	95.8
gemini-3-flash (Team and DeepMind, 2024)	99.4	94.6	90.3	90.7	95.2	95.3
deepseek-chat (DeepSeek-AI, 2024)	99.4	97.6	92.3	92.5	96.6	96.6
LDTL	98.9	95.4	78.8	87.9	93.4	91.7

We further compare our method with several widely used commercial LLMs, including GPT-4o, GPT-4o-mini, Claude-Sonnet-4, Gemini-3-Flash, and DeepSeek-Chat. These models are used purely for inference within the same sequential diagnostic framework. At each step, the model predicts the next diagnostic action and updates the disease prediction based on the currently available evidence. In this setting, the planning stage is implemented as direct next action prediction by the LLM, without latent path modeling or trajectory-level regularization. As shown in Table 3, these models achieve strong predictive performance, with mean accuracy between 95.2 and 96.6 and macro F1 scores around 95.

Our method LDTL achieves a mean accuracy of 93.4 and a macro F1 score of 91.7. While slightly lower than the strongest proprietary models, it is important to note that these commercial systems typically

rely on substantially larger model scales and extensive pretraining corpora. As a result, they may benefit from stronger general medical knowledge and prior exposure to similar clinical descriptions. In contrast, LDTL is built on a significantly smaller open source model llama-3-8b and trained for this sequential diagnostic task. Despite this difference in model capacity, our approach achieves comparable performance, highlighting the effectiveness of the proposed framework.

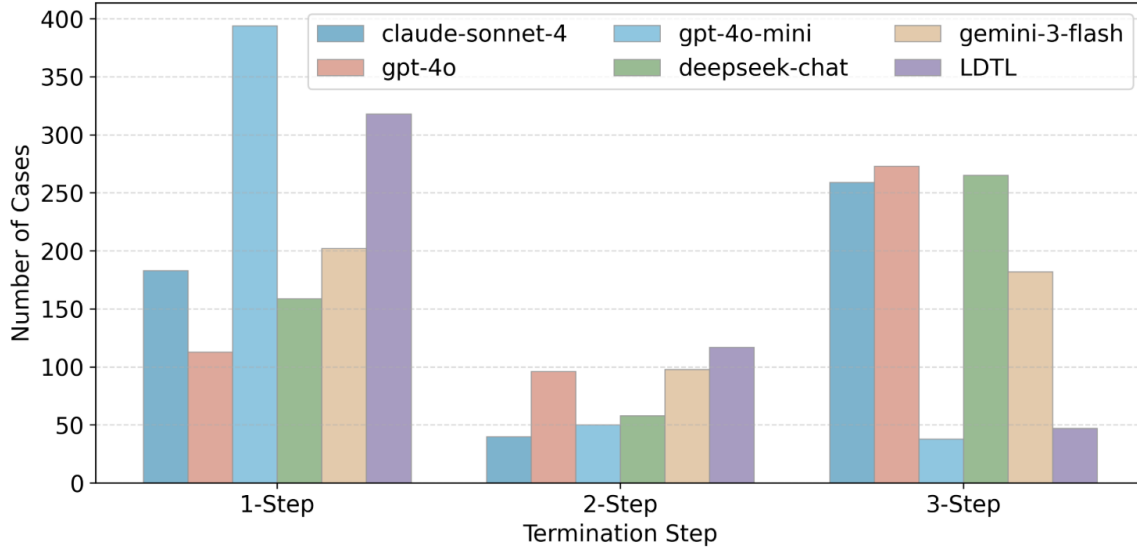


Figure 5: Distribution of diagnostic termination steps on the test cases. Numbers indicate how many cases terminate after one, two, or three actions.

Figure 5 compares the distribution of termination steps across different models. Our method LDTL frequently reaches a diagnosis in early stages, with a large portion of cases terminating after a single step with 318 cases, suggesting that the model is able to identify informative evidence early in the diagnostic process. At the same time, it continues to request additional examinations when uncertainty remains, resulting in a moderate number of two-step decisions. In contrast, several strong LLM baselines such as GPT-4o, Claude-Sonnet-4, and DeepSeek-Chat tend to rely more heavily on three-step diagnostic sequences, indicating a more conservative strategy that gathers additional evidence before committing to a diagnosis. Gemini-3-Flash shows a more balanced distribution across one, two, and three-step terminations. Besides, GPT-4o-mini shows a strong bias toward one-step termination, suggesting a preference to make early decisions with minimal evidence.

Overall, these patterns highlight differences in diagnostic behavior across models, where LDTL demonstrates a more efficient evidence acquisition strategy while still allowing additional diagnostic steps when necessary.